

SAMUEL HILL, Australia

WMO: 943700

Lat: 22.743S

Lon: 150.658E

Elev: 32

StdP: 100.95

Time Zone: 10.00 (E10)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.0	5.7	0.2	3.9	12.9	2.5	4.5	11.6	7.2	19.9	6.6	19.9	0.4	170	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	32.4	24.0	31.4	23.7	30.4	23.3	26.0	29.0	25.4	28.4	24.9	27.9	4.7	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.4	27.1	24.7	19.7	26.7	24.1	19.1	26.3	80.8	29.2	78.4	28.2	76.3	28.1	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.6	6.2	DB	1.9	34.6	1.8	1.3	0.6	35.6	-0.5	36.3	-1.5	37.1	-2.9	38.0	(4)
(5)			WB	1.4	26.8	1.8	0.7	0.2	27.3	-0.9	27.7	-1.9	28.0	-3.1	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	25.8	25.7	24.7	22.3	19.0	16.8	16.0	16.2	18.7	21.2	23.1	25.0	(6)	
	DBStd	4.13	1.45	1.32	1.44	1.49	2.35	2.69	2.72	2.42	2.13	2.14	1.82	1.65	(7)	
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD18.3	254	0	0	0	1	21	59	79	73	19	2	0	0	(9)	
	CDD10.0	4085	490	441	454	369	279	204	188	193	261	347	393	465	(10)	
	CDD18.3	1297	232	208	196	119	42	14	8	7	31	91	143	207	(11)	
	CDH23.3	9618	2030	1707	1319	539	133	27	24	66	278	632	1109	1754	(12)	
(13)	CDH26.7	2406	621	476	254	52	5	0	0	2	27	114	289	566	(13)	
(14)	Wind	WSAvg	2.6	3.2	3.2	3.3	2.8	2.3	2.1	2.1	2.0	2.2	2.7	2.8	3.0	(14)
(15) Precipitation	PrecAvg	1406	224	288	191	116	91	82	70	39	38	48	77	143	(15)	
	PrecMax	2687	622	743	438	195	269	214	285	136	140	111	233	612	(16)	
	PrecMin	823	27	51	47	18	5	4	5	0	1	2	20	7	(17)	
	PrecStd	477	154	220	131	48	61	59	77	41	39	30	64	132	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	32.9	31.7	29.7	27.3	25.4	25.4	26.5	28.8	30.5	32.2	33.7	(19)	
		MCWB	24.4	25.0	24.7	22.6	21.0	19.3	18.4	19.2	20.1	20.7	22.2	23.1	(20)	
	2%	DB	32.1	31.7	30.2	28.0	26.0	24.0	23.9	25.0	27.5	29.1	30.6	32.2	(21)	
		MCWB	24.4	24.4	23.9	21.5	19.7	18.7	17.8	17.7	19.5	21.3	22.2	23.4	(22)	
	5%	DB	31.0	30.6	29.1	27.1	24.8	22.9	22.7	23.9	26.2	28.0	29.5	31.0	(23)	
		MCWB	23.9	24.0	23.4	21.4	19.3	18.3	17.4	17.4	18.9	20.9	21.7	23.1	(24)	
	10%	DB	29.8	29.6	28.1	26.2	23.7	21.8	21.5	22.7	25.0	26.8	28.4	29.7	(25)	
		MCWB	23.5	23.8	22.9	21.3	19.2	17.9	17.1	16.9	18.2	20.2	21.3	22.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.8	26.1	24.9	22.9	21.6	21.4	21.4	22.5	23.6	25.1	26.1	(27)	
		MCDB	29.6	29.5	29.3	27.2	25.2	23.2	22.9	23.9	25.7	27.0	28.4	29.5	(28)	
	2%	WB	25.8	26.0	25.3	23.7	21.7	20.5	20.0	20.1	21.4	22.8	23.8	25.2	(29)	
		MCDB	29.1	28.7	28.1	25.7	24.2	22.2	22.2	23.4	24.5	26.6	27.6	28.5	(30)	
	5%	WB	25.2	25.4	24.7	22.8	20.8	19.7	19.0	18.9	20.6	22.0	23.1	24.6	(31)	
		MCDB	28.3	28.2	27.4	25.0	23.3	21.5	20.9	21.8	23.9	26.0	27.2	27.9	(32)	
	10%	WB	24.7	24.9	24.1	22.2	20.1	18.9	18.1	18.0	19.7	21.3	22.4	23.9	(33)	
		MCDB	27.8	27.7	26.5	24.8	22.5	20.9	20.2	21.1	23.4	25.1	26.6	27.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	7.0	6.7	7.9	9.8	10.2	11.5	13.4	12.7	11.0	10.2	9.0	(35)	
		MCDBR	10.3	9.8	8.6	10.2	11.4	11.4	12.6	15.0	14.4	12.5	12.3	11.9	(36)	
	5% WB	MCWBR	3.8	3.8	3.6	5.0	6.1	6.4	7.2	8.5	7.4	5.7	5.2	4.4	(37)	
		MCWBR	7.9	7.5	6.6	7.3	8.4	7.7	8.7	11.7	11.2	10.3	10.1	8.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.400	0.394	0.380	0.357	0.336	0.322	0.313	0.323	0.354	0.380	0.396	0.398	(40)	
		taud	2.506	2.530	2.556	2.600	2.609	2.658	2.653	2.609	2.509	2.471	2.454	2.487	(41)	
	Ebn at Noon	Ebn at Noon	945	937	924	905	886	883	903	927	934	940	942	948	(42)	
		Edh at Noon	115	110	103	91	83	76	78	89	106	116	120	117	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.54	6.04	5.50	4.76	4.09	3.57	4.04	4.91	5.98	6.61	6.89	6.85	(44)	
	RadStd	RadStd	0.42	0.66	0.45	0.41	0.22	0.32	0.30	0.33	0.41	0.40	0.69	0.67	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page